

JAYANT TOMAR

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EDUCATION

Delhi Technological University (DTU), New Delhi, India

CGPA: 9.34

Bachelor of Technology in Computer Science and Engineering

2024 – 2028

Class 12th CBSE Board

Percentage: 95%

Class 10th CBSE Board

Percentage: 97.2%

TECHNICAL SKILLS

Languages: C/C++ (C++17), Python, SQL, HTML, CSS

Backend & Systems: REST APIs, TCP/IP, WebSockets, Multithreading, POSIX, Vector Search, Ollama

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

Developer Tools: Git/GitHub, Vercel, VS Code, Visual Studio, MySQL, Google Colab, Kaggle, JupyterLab

PROJECTS

Fast Hybrid Search Engine | C++, BM25, Vector Search, Trie, Multithreading

[GitHub](#) | [Video](#)

- Built a **multithreaded C++ Hybrid Search Engine** merging exact keyword matching (**BM25 Inverted Index**) with local AI **vector embeddings (Ollama)** for semantic retrieval.
- Enhanced search accuracy and latency by implementing **Trie-based autocomplete**, **Levenshtein spell correction**, a thread-safe **LRU query cache**, and an **Top-K Min-Heap** for pagination.
- Designed and implemented a C++ HTTP backend exposing **REST APIs** for search, snippets, autocomplete, and benchmarking, with **CORS preflight handling** and **PDF binary extraction**.
- Built a **hybrid indexing pipeline** with incremental ingestion, **POSIX mmap** persistence, and **startup garbage collection** for zero-latency restarts.
- Implemented **lock-free parallel document indexing** using **thread-local inverted indexes** with **post-merge** aggregation, achieving **2.14×** indexing speedup and average search latency (**0.75 ms**, **1319 queries/sec** throughput).

High-Performance In-Memory Key-Value Store | C++, TCP

[GitHub](#) | [Video](#)

- Engineered **thread-safe** operations using **mutex-based** synchronization, ensuring **data integrity** for over 500 concurrent clients with sub-millisecond latency (**0.305 ms** avg, $p99 < 0.9ms$).
- Architected **hybrid persistence (WAL + snapshot/RDB)** enabling crash recovery and fast restart by combining log replay with periodic full-state snapshots.
- Streamlined **LRU cache eviction with heap-optimized TTL expiration ($O(\log n)$)**, replacing full scans, sustaining 55K TTL writes/sec with only **0.018 ms** avg latency..
- Developed **master-slave replication with initial synchronization**, where slaves receive snapshot-based state transfer followed by real-time command replication for consistency.
- Achieved **93K+** GET ops/sec and **60K+** SET ops/sec while integrating graceful shutdown and structured logging, and a zero-dependency **WebSocket-to-TCP proxy bridge** for real-time visual telemetry and cluster state exploration.

ACHIEVEMENTS AND AWARDS

- **Competitive Programming: Knight** (1885) at LeetCode, **Pupil** at Codeforces, and **3-Star** at CodeChef.
- **Algorithmic Consistency:** Solved **650+** **DSA problems** on LeetCode, maintaining a rigorous 200-day streak.
- **HPAIR Asia Conference 2026:** Selected as a global delegate for the Harvard College Project for Asian and International Relations (HPAIR) Conference in VinUniversity in Hanoi, Vietnam. - [Acceptance Letter](#)
- **Top 12 Finalist:** Hack-LLM (24-Hour Hackathon), IIIT Delhi - [Certificate](#)
- **Goldman Sachs India Hackathon 2026-CS:** Secured All India Rank **1127** - [Leaderboard](#)
- **JEE Advanced & Mains 2024:** Secured **AIR 10,987** in JEE Advanced and **99.27** percentile in JEE Mains.

LEADERSHIP & CO-CURRICULARS

UGV-DTU (Unmanned Ground Vehicle Team)

Sept 2024 – May 2025

- Collaborated with the team in testing the ZED 2i camera to evaluate its capabilities against IGVC 2025 requirements.

CERTIFICATIONS

Stanford Algorithms Specialization
Kaggle Intermediate ML

Machine Learning Specialization (Andrew Ng)
Kaggle Intro to ML